CPSC 536C: Algorithms for Convex Optimization
University of British Columbia

Gradient Descent

Akshay Ram
February 2, 2025

These notes are subject to change and may contain errors.

1 Introduction

The goal of this note is to study and analyze the following simple procedure for unconstrained convex optimization:

$$x_{t+1} := x_t - \eta_t \nabla f(x_t).$$

Here and in the remainder we assume f is differentiable (so $\nabla f(x)$ is well-defined). We note that much of the analysis carries over to the non-differentiable convex functions by using any sub-gradient.

In contrast to cutting plane methods, these gradient methods usually have $\text{poly}(1/\varepsilon)$ convergence to an ε-approximate solution, but have much better dependence on the problem dimension n. At the end we will also show that such dependencies are necessary by constructing and analyzing certain hard instances for gradient descent.

1.1 Approximate Solutions

We consider more formally what it means to produce an approximate solution:

Definition 1. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable convex function with optimizer $x^* := \arg\min_x f(x)$.*

- *x has ε-function gap if $f(x) \le f(x^*) + \varepsilon$;*
- *x is ε-close if $\|x - x^*\|_2 \le \varepsilon$;*
- *x is ε-critical if $\|\nabla f(x)\|_2 \le \varepsilon$.*

These three notions of approximate solutions are in general incomparable, and their importance depends on the application at hand.

Exercise 1. *Compute the function gap, distance to opt, and gradient for the following:*

1. *$f(x) := \|x\|_2^2/2$;*
2. *$f(x) := \|Ax - b\|_2^2$;*
3. *$f(Q) := -\log\det(Q) + \sum_i \lambda_i \langle x_i, Qx_i\rangle$.*

1.2 Assumptions

We need some assumptions on our initial iterate and the function in order to guarantee any reasonable approximation. The following are natural assumptions satisfied by many functions in practice.

Definition 2. *$f : \mathbb{R}^n \to \mathbb{R}$ is G-Lipschitz if*

$$\forall x, y : \quad |f(y) - f(x)| \leq G\|y - x\|_2.$$

This intuitively allows us to relate f to an affine function with slope G. We can also relate f to quadratic functions, which are the next simplest convex functions. Recall the first-order definition of convexity:

$$\forall x, y : \quad f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

Definition 3. *$f : \mathbb{R}^n \to \mathbb{R}$ is α-strongly convex if*

$$\forall x, y : \quad f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\alpha}{2}\|y - x\|_2^2.$$

f is β-smooth if

$$\forall x, y : \quad f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2}\|y - x\|_2^2.$$

Note that the first gives a stronger lower bound than standard convexity, whereas the smoothness condition gives an upper bound. In both cases this allows us to compare our function to a convex quadratic, which is much simpler to analyze.

Claim 4. *For quadratic $q_\gamma(x) := q_0 + \langle b, x - x_0 \rangle + \frac{\gamma}{2}\|x - x_0\|_2^2$, we can optimize*

$$\min_x q_\gamma(x) = q_0 - \frac{\|b\|_2^2}{2\gamma} \quad \textit{with} \quad x^* := \arg\min_x q_\gamma(x) = x_0 - \frac{b}{\gamma}.$$

We leave the proof as an exercise (show directly by computing gradient and setting to 0).

Exercise 2.

1. *If f is α-strongly convex, then it is also α'-strongly convex for any $\alpha' \leq \alpha$. I.e. strong convexity is an increasingly strong assumption for α increasing;*
2. *If f is β-smooth, then it is also β'-strongly convex for any $\beta' \geq \beta$. I.e. smoothness is a weaker assumption for β increasing;*
3. *If f, g are α_f, α_g-strongly convex and β_f, β_g-smooth, respectively, then $f+g$ is $\alpha_f+\alpha_g$-strongly convex and $\beta_f + \beta_g$-smooth;*

Exercise 3. *Twice-differentiable f is β-smooth iff the gradient is β-Lipschitz:*

$$\forall x, y : \quad \|\nabla f(x) - \nabla f(y)\|_2 \leq \beta\|x - y\|_2.$$

The above gives a bit of intuition as to the usefulness of smoothness: gradient descent takes the locally steepest direction of improvement; so if the gradient changes slowly, then this remains a good direction for some bounded distance, guaranteeing significant improvement in each step.

The point of the above assumptions are that they allow us to relate various notions of approximate solutions.

Fact 5. *Let x^* be the optimizer of f, which is (1) G-Lipschitz, (2) β-smooth, and (3) α-strongly convex, respectively. Then*

1. *$f(x) - f(x^*) \leq G\|x - x^*\|_2$;*
2. *$f(x) - f(x^*) \leq \frac{\beta}{2}\|x - x^*\|_2^2$;*
3. *$f(x) - f(x^*) \leq \frac{\|\nabla f(x)\|_2^2}{2\alpha}$ and $f(x) - f(x^*) \geq \frac{\alpha}{2}\|x - x^*\|_2^2$.*

2 Analysis of Gradient Descent

Intuitively, if we have a simple quadratic function, theorem 4 shows how to find the optimum directly. In general, convexity (and strong convexity and smoothness) allows us to approximate our function f by an affine or quadratic function. The algorithm and analysis of gradient involves using these proxies to compute updates and reason about the progress made.

Theorem 6. *Let f be α-strongly convex and β-smooth, and consider update*

$$x_{t+1} := x_t - \frac{1}{\beta}\nabla f(x_t).$$

Then the function gap of iteration T can be bounded as

$$f(x_T) - f(x^*) \leq \left(1 - \frac{\alpha}{\beta}\right)^T (f(x_0) - f(x^*)),$$

*i.e. we achieve ε-*relative *function gap in $O(\kappa \log(1/\varepsilon))$ iterations, where $\kappa := \beta/\alpha$.*

Proof: The update step can be derived as the optimizer of the proxy given by smoothness:

$$x_{t+1} = \arg\min_x q_\beta(x) := \arg\min_x f(x_t) + \langle \nabla f(x_t), x - x_t\rangle + \frac{\beta}{2}\|x - x_t\|_2^2,$$

where the minimum calculation follows by theorem 4. By smoothness we have $f \leq q_\beta$, so we can show significant progress in function value:

$$f(x_{t+1}) \leq q_\beta(x_{t+1}) = f(x_t) - \frac{\|\nabla f(x_t)\|_2^2}{2\beta},$$

where again the last step was by theorem 4. Next, we use strong convexity to show that this gives a significant improvement:

$$\frac{\|\nabla f(x_t)\|_2^2}{2\alpha} \geq f(x_t) - f(x^*),$$

where we applied theorem 5(3) for α-strongly convex f. Putting this together gives

$$\begin{aligned} f(x_{t+1}) - f(x^*) &= (f(x_{t+1}) - f(x_t)) + (f(x_t) - f(x^*)) \\ &\leq -\frac{\|\nabla f(x_t)\|_2^2}{2\beta} + (f(x_t) - f(x^*)) \leq (1 - \alpha/\beta)(f(x_t) - f(x^*)), \end{aligned}$$

where in the second step we used the upper bound for the update, and in the final step we used that theorem 5(3) to compare the gradient norm and $f(x_t) - f(x^*)$. □

Note that the above gives exponential convergence in terms of function gap. Also by theorem 5 we can show convergence in terms of the solution $\|x_t - x^*\|_2$ and gradient $\|\nabla f(x_t)\|_2$. The above setting is in some sense the most structured, since we have upper and lower bounds on f via smoothness and strong convexity. In the next settings, we remove the strong convexity assumption.

Theorem 7. *Let f be β-smooth and assume $\|x_0 - x^*\|_2 \leq R$. Then for update*

$$x_{t+1} := x_t - \frac{\nabla f(x_t)}{\beta},$$

the function gap can be bounded as

$$f(x_T) - f(x^*) = \min_{t\in[T]} f(x_t) - f(x^*) \lesssim \frac{\beta R^2}{T},$$

*i.e. we achieve ε-*relative *function gap in $O(1/\varepsilon)$ iterations.*

Proof: By theorem 4 we argue that

$$x_{t+1} = \arg\min_x q_\beta(x) := \arg\min_x f(x_t) + \langle \nabla f(x_t), x - x_t \rangle + \frac{\beta}{2}\|x - x_t\|_2^2.$$

By the smoothness condition, this guarantees improvement in the objective

$$f(x_{t+1}) \le q_\beta(x_{t+1}) = f(x_t) + \langle \nabla f(x_t), x_{t+1} - x_t \rangle + \frac{\beta}{2}\|x_{t+1} - x_t\|_2^2 = f(x_t) - \frac{\|\nabla f(x_t)\|_2^2}{2\beta}.$$

In order to argue about optimality gap, we use β-strong convexity of q_β: for arbitrary x we have

$$\begin{aligned} f(x) + \frac{\beta}{2}\|x - x_t\|_2^2 &\ge f(x_t) + \langle \nabla f(x_t), x - x_t \rangle + \frac{\beta}{2}\|x - x_t\|_2^2 = q_\beta(x) \\ &\ge q_\beta(x_{t+1}) + \langle \nabla q_\beta(x_{t+1}), x - x_{t+1} \rangle + \frac{\beta}{2}\|x - x_{t+1}\|_2^2 \\ &\ge f(x_{t+1}) + \frac{\beta}{2}\|x - x_{t+1}\|_2^2, \end{aligned}$$

where the first step was by convexity of f, the third step follows by β-strong convexity of q_β (as q_β is the sum of an affine function and $\frac{\beta}{2}\|x^* - x_t\|_2^2$), along with the fact that x_{t+1} is the optimizer of q_β so the gradient term vanishes, and in the last step we used β-smoothness condition to upper bound $f(x_{t+1})$. Plugging in $x = x^*$ above and rearranging gives

$$f(x_{t+1}) - f(x^*) \le \frac{\beta}{2}(\|x^* - x_t\|_2^2 - \|x^* - x_{t+1}\|_2^2).$$

Then we can apply a telescoping argument to show

$$\frac{1}{T}\sum_{t=0}^{T-1}(f(x_{t+1}) - f(x^*)) \le \frac{1}{T}\sum_{t=0}^{T-1}\frac{\beta}{2}(\|x^* - x_t\|_2^2 - \|x^* - x_{t+1}\|_2^2) \le \frac{\beta\|x_0 - x^*\|_2^2}{T},$$

where in the last step we used telescoping and $\|x_T - x^*\|_2 \ge 0$. Finally, to argue about the last iterate, we note that plugging in $x = x_t$ into the bound above gives

$$f(x_t) = f(x_t) + \frac{\beta}{2}\|x_t - x_t\|_2^2 \ge f(x_{t+1}) + \frac{\beta}{2}\|x_t - x_{t+1}\|_2^2 \ge f(x_{t+1}),$$

i.e. the function value is non-increasing, and therefore the last iteration is minimum. □

The non-smooth setting is the least structured, and will have worse guarantees.

Theorem 8. *Let f be G-Lipschitz and assume $\|x_0 - x^*\|_2 \le R$. Then for update*

$$x_{t+1} := x_t - \eta\nabla f(x_t) \quad \textit{where} \quad \eta := \sqrt{\frac{R^2}{G^2 T}},$$

the function gap can be bounded as

$$\min_{t\in[T]} f(x_t) - f(x^*) \lesssim \sqrt{\frac{R^2 G^2}{T}},$$

*i.e. we achieve ε-*relative *function gap in $O(1/\varepsilon^2)$ iterations.*

In the next section we show that the guarantee for the Lipschitz setting is optimal! I.e. there exists worst-case instances such that *any* algorithm with access to function evaluation and gradients must have iteration dependence of the form $1/\varepsilon^2$. On the other hand, the guarantee for the first two cases, those involving smoothness, are surprisingly not optimal.